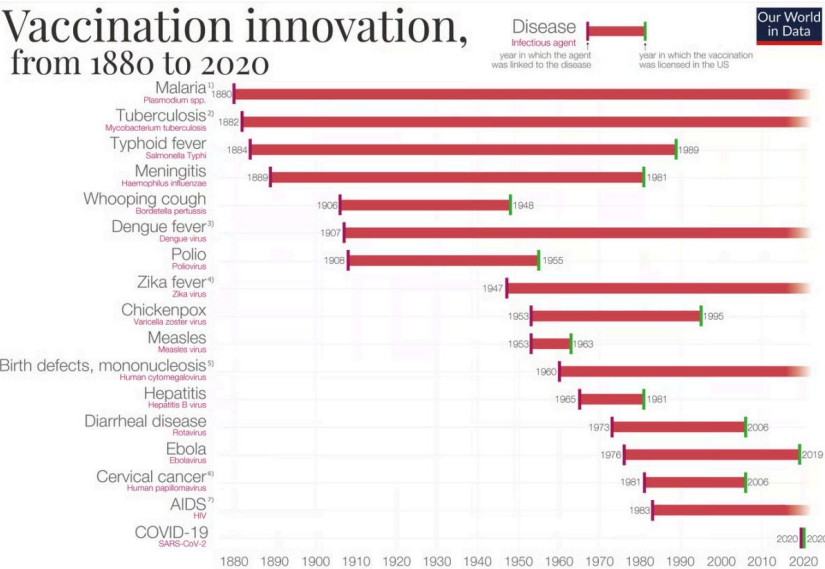
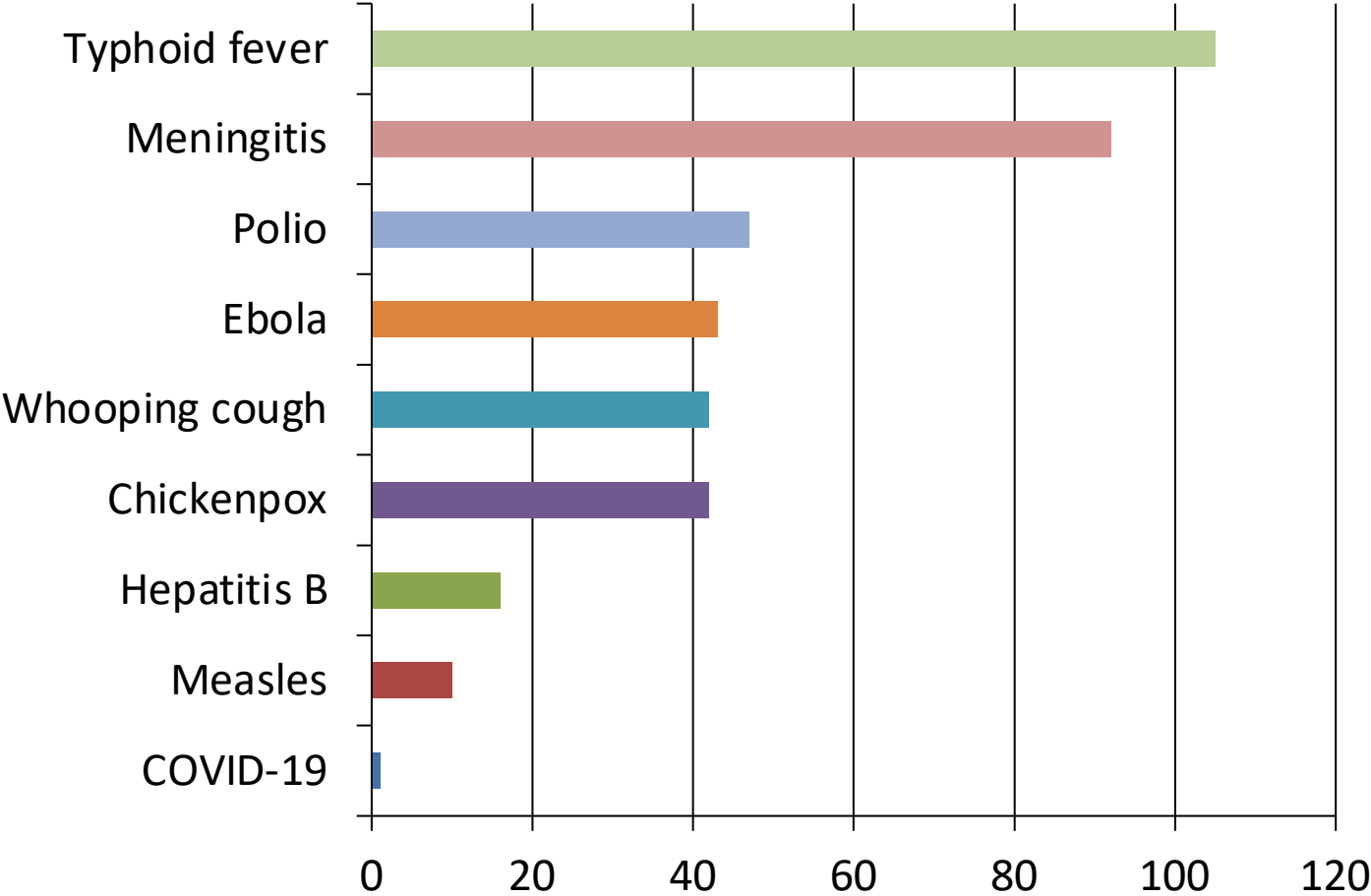


The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of how vaccines have transformed public health (1796–2020).
Source: Our World in Data

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Years from Identification to Licensure



100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-Vaccine Cases\n(per mil/yr)	Post-Vaccine Cases\n(per mil/yr)	Case Reduction	Pre-Vaccine Deaths\n(per mil/yr)	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Measles	3,044	0.2	99.99%	2.5	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	100%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Vaccines reduced measles cases across US states



Data source: Project Tycho (2018); Centers for Disease Control and Prevention (1959–2022)

OurWorldinData.org — Research and data to make progress against the world's largest problems.

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Milestones:

1963: First measles vaccine developed by John Enders

1971: MMR vaccine developed by Maurice Hilleman

1980: Vaccination became mandatory for children entering kindergarten

Before 1963: 100–1,000+ cases per 100,000.

After 1980: Near zero across all states.

Smallpox: The Only Human Disease Eradicated Through Vaccination

- Smallpox was declared eradicated worldwide in 1980.
- 1959: WHO passes a resolution to eradicate smallpox.
- 1967: WHO launches an intensified global campaign.
- 1977: Last natural case occurs in Somalia.
- 1980: Smallpox eradication is certified globally.
- **Key Enablers:**
 - Virus infected only humans (no animal reservoirs).
 - Symptoms were easily distinguishable.
 - The vaccine was cheap, stable, and highly effective.

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

- Vaccines protect not only individuals but entire communities.
- When enough people are vaccinated, pathogens cannot spread effectively.
- This indirect protection shields vulnerable individuals who cannot be vaccinated:
 - Newborns too young for vaccines.
 - Immunocompromised individuals.
 - Cancer patients undergoing chemotherapy.
- Maintaining high vaccination rates prevents outbreaks and protects the most vulnerable.



Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

- Despite incredible successes, significant challenges remain in the fight against vaccine-preventable diseases.
- **Incomplete control of certain diseases in the US:**
 - Pneumococcal disease: 40.5% case reduction, 31.3% death reduction.
 - Hepatitis A: 89% case reduction.
 - Acute hepatitis B: 83.9% case reduction.
- **Global and systemic challenges:**
 - Access gaps: Not everyone globally has access to life-saving vaccines.
 - Supply shortages: Disruptions put children at risk.
 - Trust challenges: Poor scientific understanding can reduce vaccine trust and uptake.

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- 100% elimination achieved for diphtheria, smallpox, and polio in the US.
- Development timelines collapsed from over 100 years to under 1 year, driven by mRNA and genomics.
- Herd immunity is critical to protecting vulnerable populations like newborns and cancer patients.
- Coverage gaps persist for diseases such as pneumococcal disease, hepatitis A, and hepatitis B.
- **Call to Action: Global health systems must focus on improving vaccine access, addressing supply shortages, and building public trust to finish the job.**